* Variants of Naive Baye's

1. Bernoulli Naive Baye's
2. Multinomial Naive Baye's
3. Gaussian Naive Baye's

1. Bernoulli Naive Bayes:

Whenever your feature are following a Bernoulli Distribution Then we need to use Bernoulli Naive Baye's Algorithm.

Data set

Bernoulli → 0, 1

f_1	f_2	f_3	o/p
Yes	Pass	Male	Yes
Yes	fail	female	No
No	Pass	Male	Yes
Yes	fail	female	NO

2. Multinomial Naive Bayes ⇒ i/p → Text

Dataset : Spam Classification

i/p → Email Body	Spam/Not Spam ← o/p
You have won million $$ lottery	Spam
Hey you have done a good Job.	HAM ← Not Spam

↓

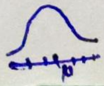
Numerical values
↳ vectors.

Natural Language Processing

1. BOW
2. TF - IDF
3. word2vec

3. Gaussian Naive Baye's

If the features are following Gaussian Distribution, Then we are Gaussian Naive Bayes



DATASET → continuous

Age	Height	weight	Yes/No
25	170	78	
38	180	75	
22	150	60	
34	140	35	